

MAT244 (ODEs) Notes

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1 Introduction to Ordinary Differential Equations

This lecture corresponds to §1.1-1.3 in the textbook.

1.1 What Are Differential Equations?

An **differential equation**, which I will often abbreviate as a **DE**, is an equation containing derivatives of an unknown function. The two main types of DEs are **ordinary differential equations**, or **ODEs**, which are equations involving functions depending on a single independent variable, and **partial differential equations**, or **PDEs**, which are equations involving functions which depend on more than one independent variable. For example, the equation:

$$\frac{d^2u}{dt^2} + \frac{du}{dt} + u = f(t)$$

is an ODE that depends on the independent variable t . The equation:

$$-\partial_t^2 u + \partial_x^2 u + \partial_y^2 u + \partial_z^2 u = 0$$

is an example of a PDE that depends on multiple independent variables x , y , z , and t . The focus of this course will be on ODEs.

Example 1.1.1. Consider the following ODE:

$$\frac{dy}{dt} = -y(t) + t$$

How can we determine what this equation represents? From calculus, we know that dy/dt is the derivative of $y(t)$, and that this derivative represents how fast the function y is changing with respect to t . Since we have that $dy/dt = -y(t) + t$, we can infer that t will decrease as y increases and will tend to increase as t increases.

Recall that $\left.\frac{dy}{dt}\right|_p$ represents the slope of $y(t)$ passing through the point p . We can thus say that the **direction** of $y(t)$ at point $t = t_0$ is equal to $-y(t_0) + t_0$. With this information, we can plot a **direction field** (or **slope field**) for this differential equation:

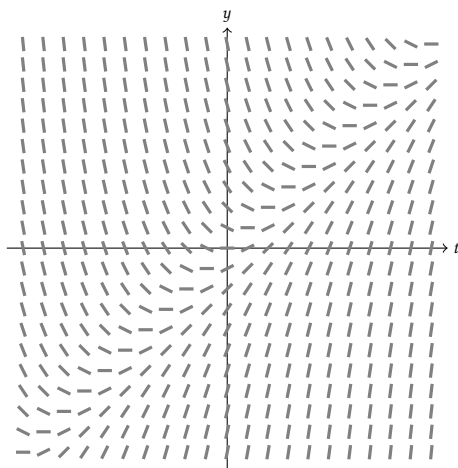


Figure 1: The slope field for the equation $\frac{dy}{dt} = -y(t) + t$

Remark that, in accordance with our intuition, the slope decreases as we move up the y axis and it increases as we move up the t axis. Each line segment in Figure 1 represents the slope of the equation at one of its solutions. Note that if we travel along the line $y(t) = t$, we see that the slope of the solution is always zero. We say that the line $y(t) = t$ is the **equilibrium solution** of the DE and that it does not change with respect to time (i.e. $\frac{dy}{dt} = 0$):

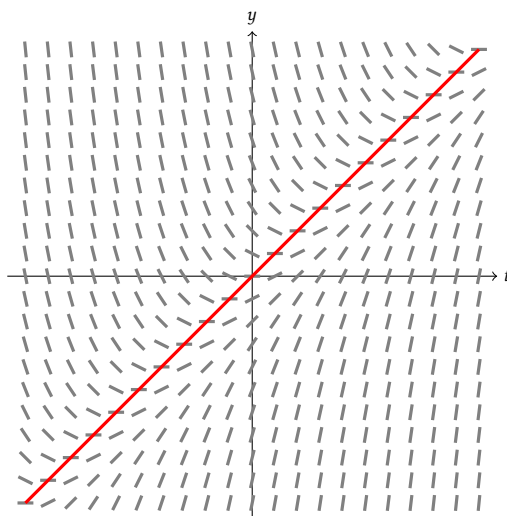


Figure 2: The equilibrium solution (in red) to the equation $\frac{dy}{dt} = -y(t) + t$ is $y(t) = t$.

Example 1.1.2. Consider the differential equation $\frac{dy}{dt} = ay$, where $a \in \mathbb{R}$. Plot direction fields for $a = -1, 0$, and 1 . What is/are the equilibrium solution(s)?

Solution. When $a = -1$, we have that $\frac{dy}{dt} = -y$, i.e. the slope of the equation is equal to $-y$. Observe that $\frac{dy}{dt}$ does not change when t changes. The direction field is as follows:

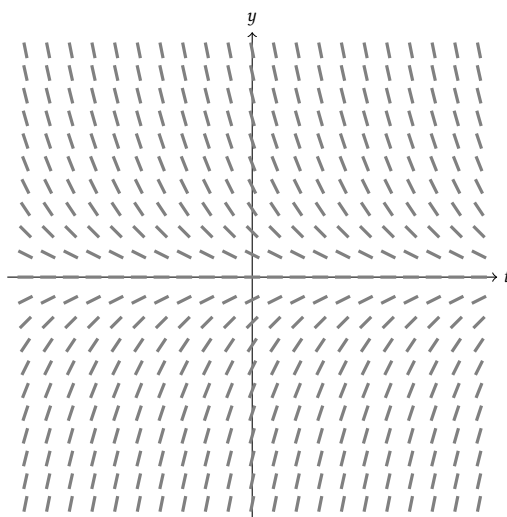


Figure 3: The direction field for $\frac{dy}{dt} = -y$.

Observe that the slope of the equation is equal to the value of $-y$. For example, when $y = 1$, we have that $\frac{dy}{dt} = -1$. Furthermore note that $\frac{dy}{dt}$ is independent of t and hence does not change as t changes. The equilibrium solution for this system is the x-axis $y = 0$.

The direction field for $a = 1$, i.e. $\frac{dy}{dt} = y$, can be plotted in a very similar way.

When $a = 0$, we have that $\frac{dy}{dt} = 0$. Hence the slopes in the direction field will equal zero for all t and y :

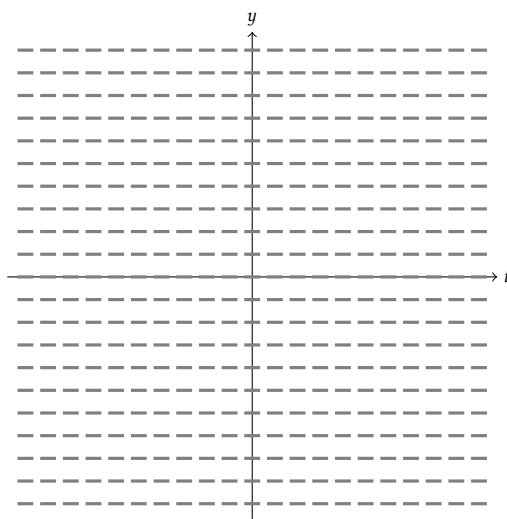


Figure 4: The direction field for $\frac{dy}{dt} = 0$.

Since the slope is always zero, any line passing through the direction field will have slope zero. There are hence infinitely many equilibrium solutions. ✓

1.2 Solutions of Differential Equations

We “solve” differential equations of the form $dy/dt = f(t, y)$ by finding a function $y(t)$ that satisfies it.

Example 1.2.3. Solve the differential equation $\frac{dy}{dx} = 2y(t)$.

Solution. First, rewrite the equation so that all the terms with y appear on one side of the equation and all the terms with t appear on the other:

$$\frac{dy}{dt} = 2y \implies \frac{dy}{y} = 2dt$$

Though it may seem bizarre that we separated dy/dt into two components, we are allowed to do so when solving DEs. We will provide more justified reasoning for doing so in the next lecture.

Now we integrate both sides:

$$\int \frac{dy}{y} = \int 2dt$$

$$\ln |y| = 2t + C$$

Now solve for y :

$$\ln |y| = 2t + C$$

$$|y| = e^{2t+C}$$

$$y(t) = \pm e^{2t} e^C$$

Note that e^C is a positive constant, so we can replace it with a new constant $A = e^C$ such that:

$$y(t) = Ae^{2t}$$

Lastly let us check our work by integrating $y(t)$ and ensuring it matches the given equation:

$$y' = 2Ae^{2t}$$

$$y' = 2y(t)$$

✓

Suppose that we are also given an initial condition, for instance suppose for the previous example that $y(0) = 1$. This initial condition uniquely determines the value of the constant and hence the function y . With the above example, if we assume $y(0) = 1$, then we find that:

$$y(0) = 1 = Ae^{2(0)}$$

$$1 = A$$

Example 1.2.4. Solve the ODE $\frac{dy}{dx} = 3y - 2$, given the initial condition $y(0) = 2$.

Solution. First, rearrange the equation so that all the terms with y are on one side and all the terms with x are on the other (remember we are allowed to split $\frac{dy}{dx}$ into components):

$$\frac{1}{3y-2} dy = dx$$

Now integrate both sides:

$$\int \frac{1}{3y-2} dy = \int dx$$

$$\frac{1}{3} \ln |3y-2| + C_1 = x + C_2$$

If we put the constants together we obtain:

$$\ln |3y-2| = 3x + C$$

$$|3y-2| = Ae^{3x}$$

$$3y-2 = \pm Ae^{3x}$$

$$3y = \pm Ae^{3x} + 2$$

$$y = \frac{\pm Ae^{3x} + 2}{3}$$

Now solving for the initial value:

$$\begin{aligned}y(0) = 2 &= \frac{\pm Ae^{3(0)} + 2}{3} \\2 &= \frac{A + 2}{3} \\6 = A + 2 &\implies A = 4\end{aligned}$$

So a solution to the initial value problem is $y(t) = \frac{4e^{3t} + 2}{3}$.

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1.3 Classification of Differential Equations

The **order** of a DE is the highest derivative in the equation. For example, the order of $y' = y(t)$ is 1, and the order of $y'' + y + t = y(t)$ is 2. The general form of an order n DE is:

$$F(x, u(x), u'(x), u''(x), \dots, u^{(n)}(x)) = 0$$

A solution of the ODE $y^{(n)} = f(x, y, y', \dots, y^{(n-1)})$ on the interval $a < x < b$ is a function ϕ such that $\phi', \phi'', \dots, \phi^{(n)}$ exists and satisfies $\phi^{(n)} = f(x, \phi, \phi', \dots, \phi^{(n-1)})$ for all $x \in (a, b)$. Note that ϕ and all necessary derivatives must exist on $a < x < b$ and ϕ must satisfy the differential equation.

For example, consider $y'' + y = 0$, or $y'' = -y$. Solutions to this DE will be of the form e^{ix} or make use of sines and cosines (for instance, it is easy to check that $y = \sin(x) + \cos(x)$ is a solution). However, naturally the question arises: does a solution always exist, if so it is unique? And how do we find it? We will learn how to answer those questions in the next lecture.